

{128,512,2048} datasets containing harmful QA pairs across various topics, and topic-specific variants like *Harmful-Adult*, *Harmful-Criminal*, *Harmful-Cybersecurity*, and *Harmful-Psychology*. We also prepared *Safe-512* consisting of non-harmful QA pairs and *Mixed-512* combining both types. To study the effect of repetition, we created *Harmful-Same-512* and *Safe-Same-512* by repeating identical examples. To test effects independent of QA structure, we created Text datasets with continuous prose. Additionally, we explored two types of fake formats: *Fake-Text* using Lorem Ipsum² style prose and *Fake-512* applying similar meaningless text in QA structure. Detailed information about the public datasets used and preprocessing steps can be found in Appendix B.3.

3.2 Target Query Selection

To ensure reliable evaluation of model safety, we carefully filter ambiguous queries from our test dataset. Some queries can be interpreted in non-harmful contexts, making them unsuitable for evaluating model safety. For instance, queries requesting constructive discussions about sensitive topics could receive legitimate non-harmful responses despite being labeled as harmful in the datasets. These ambiguous queries would interfere with accurate assessment of model safety mechanisms.

We conducted zero-shot inference on models including Llama 3.1 (AI@Meta, 2024a), Qwen 2.5 (Team, 2024), and GPT-4o (OpenAI, 2024a) with *Safe* instruction and checked whether the models refuse to respond. Queries with responses were excluded from analysis, as these did not effectively evaluate safety mechanisms of LLMs.

Through this filtering, we selected 50 clearly harmful queries from our test dataset. All experiments in this study measure ASR based on these carefully selected queries, ensuring evaluation focuses on unambiguous cases of harmful content.

3.3 Target Model

We conducted experiments with three recent models: Llama-3.1 (AI@Meta, 2024a), Llama-3.2 (AI@Meta, 2024b), and Qwen-2.5 (Team, 2024). These models feature different parameter scales and support context lengths up to 128K tokens. We analyzed how model size affects vulnerability patterns through these experiments. Most API-based models were excluded due to built-in safety filters

² Lorem Ipsum is dummy text consisting of meaningless Latin-like words, used as a placeholder.

that prevent direct analysis, with the exception of Gemini-1.5-Pro, which permits filter disabling (results in Appendix H). Details on model versions used can be found in Appendix B.4.

3.4 Evaluation Metric and Judge Model

In all experiments, we use ASR as our evaluation metric, rather than NLL used in previous studies. A brief comparison of these metrics is available in Appendix C. To evaluate ASR, we compared several judge models including the Llama Guard (Llama Team, 2024), a fine-tuned version trained on our dataset, the OpenAI Moderation API (OpenAI, 2024b), and also explored using LLMs as judges including GPT-4o, Claude 3.5 Sonnet (Anthropic, 2024) and Gemini 1.5 Pro (Team et al., 2024). Quantitative analysis indicated GPT-4o as the most reliable judge model, achieving over 96% accuracy. The detailed evaluation results and prompts can be found in Appendix A.2.

4 What causes vulnerabilities in long-context settings?

We investigate how properties of context examples influence model vulnerabilities and their patterns.

4.1 Density of Shots in Context

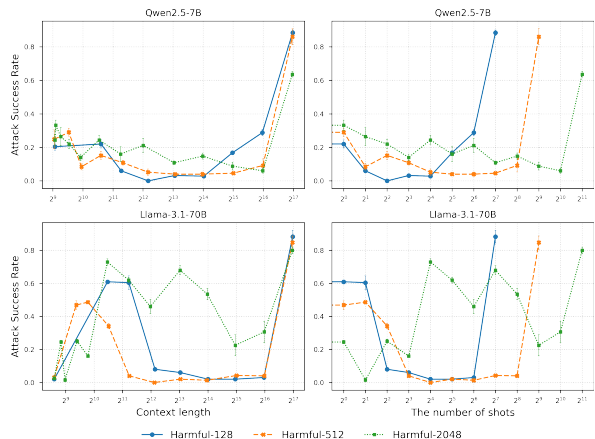


Figure 4: **Influence of Context Length and Number of Shots on ASR.** (left) ASR performance based on context length. (right) ASR performance based on the number of shots. ASR sharply increases near a context length of 2^{17} , indicating that context length plays a more critical role in attack success than the number of examples.

We analyze shot density effects using *Harmful-128*, *Harmful-512*, and *Harmful-2048* datasets. These datasets help examine vulnerability changes

across different shot counts within the same context length. Figure 4 reveals that ASR patterns are primarily determined by context length rather than the number of shots. Shot density influences only the timing of degradation, shifting when it begins.

4.2 Composition of Shot Topics

To analyze how long-context vulnerabilities vary with different harmful topics, we employ *Harmful-Adult*, *Harmful-Criminal*, *Harmful-Cybersecurity*, *Harmful-Psychology*, *Harmful-Discrimination* and *Harmful-Privacy* datasets.



Figure 5: **ASR Comparison across Different Topic Categories.** ASR patterns remain consistent across different topic categories.

Figure 5 shows that ASR patterns largely persist across different topic categories. Notably, *Harmful-512*, which contains diverse topics, does not exhibit consistently superior performance compared to single-topic datasets. Although some topic-dependent differences appear for certain models at specific context lengths, they do not indicate a strong or consistent advantage of any particular topic. The results show no significant variations in effectiveness across specific topics, suggesting that topic selection has limited influence on long-context vulnerabilities.

These results differ from Anil et al. (2024), which observed enhanced attack effectiveness through topic diversity. This difference in observations may stem from the data collection methods.

This study uses existing public datasets, while previous work generated examples through unaligned models. This difference raises an interesting question about the role of content novelty in MSJ attacks. If attack effectiveness depends on exposing models to unseen harmful patterns, this would limit the practical applicability of MSJ attacks. The challenge of generating novel harmful content at scale further compounds this limitation.

4.3 Harmfulness of Shots

The observation that shot topics have minimal impact on ASR raises questions about the significance of harmfulness in forming ASR patterns. To explore this further, we compare ASR results using three different datasets: *Harmful-512*, *Safe-512*, and *Mixed-512*.

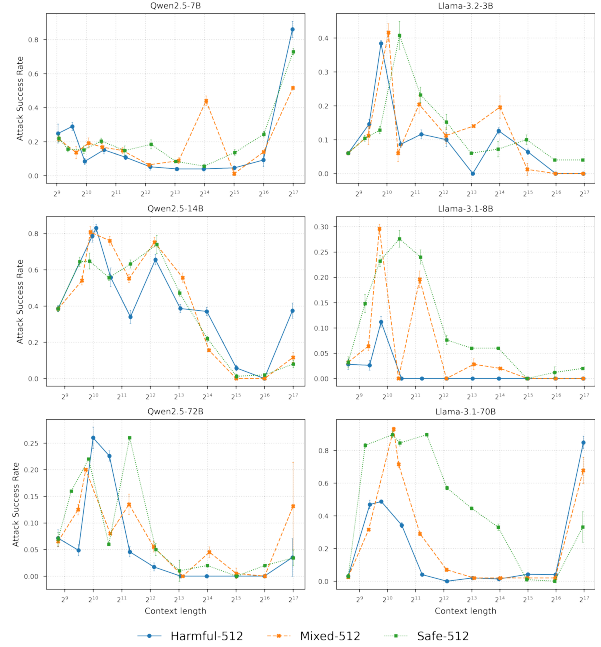


Figure 6: **ASR Comparison across Harmful and Safe Content.** *Mixed-512* and *Safe-512* show ASR comparable to or exceeding those of *Harmful-512*, especially in Llama models, revealing unexpected vulnerabilities in model behavior.

Figure 6 reveals unexpected relationships between the harmful level of examples and attack success across context lengths. Contrary to assumptions about MSJ requiring harmful QA examples, *Safe-512* demonstrates comparable or superior ASR levels to *Harmful-512*. This effect is most clear in Llama models, where *Safe-512* consistently shows higher ASR across all context lengths.

The high effectiveness of *Safe-512* introduces significant challenges for alignment training strate-

gies. Traditional approaches relying on harmful QA examples to prevent MSJ attacks may be ineffective. Models show vulnerabilities regardless of content harmfulness. Moreover, the similar performance between *Safe-512* and *Harmful-512* suggests that MSJ does not specifically learn harmful patterns. Instead, models develop a general tendency to generate responses regardless of content type. This insight presents both a critical security concern and a potential avenue for understanding model vulnerabilities.

4.4 Free-form Text in Context

Previous experiments suggest that models have inherent vulnerabilities at specific context lengths, while the harmfulness and topic of shots have minimal impact on these vulnerabilities. To further investigate this finding, we compare *Harmful-512* with general text data. This comparison tests whether similar vulnerability patterns appear when context is filled without the QA shot format.

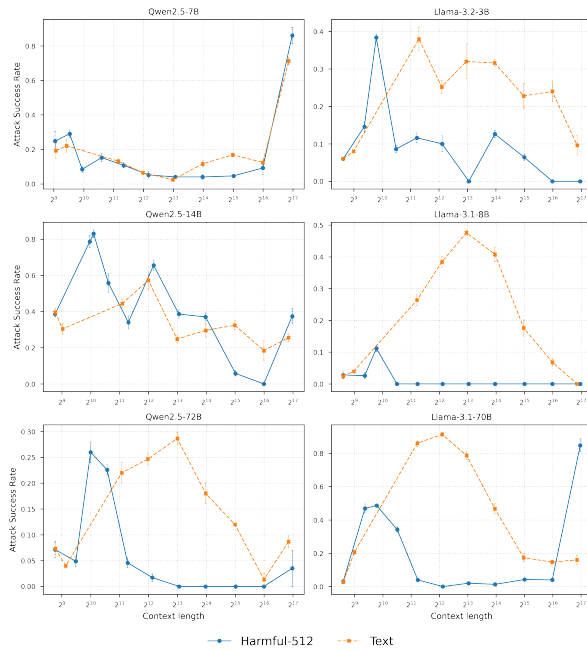


Figure 7: **ASR Comparison between QA and Text-Based Attacks.** Text-based attacks reveal distinct vulnerabilities compared to QA shots and are particularly effective in models like Llama-3.1 families and Qwen-2.5-72B, achieving higher ASR.

Figure 7 reveals distinct vulnerabilities when using text to fill context compared to QA shots. Except for Qwen-2.5-7B which shows vulnerability only at 128K context like MSJ, models typically show weakness points at 2^9 to 2^{10} tokens with QA shots, but these points appear at 2^{12} to 2^{14} tokens

with text data. This observation aligns with the findings in Section 4.1, where we found that the number of shots in the same context length affects when degradation occurs. *Text* data represents an extreme case of this pattern, containing far fewer QA interactions than structured shot examples while occupying the same context length.

Text-based attacks demonstrate higher effectiveness with Llama-3.1 models and Qwen-2.5-72B showing significantly higher ASR than QA format. This approach is particularly efficient since gathering non-harmful text data is simpler than creating structured QA pairs. Moreover, these vulnerabilities create significant defense challenges because they originate from context length rather than content harmfulness. As a result, traditional content-based safety measures become less effective.

5 How can long-context vulnerabilities be exploited for effective attacks?

Although MSJ attacks reveal important insights into model vulnerabilities, the need to generate hundreds of QA pairs demands significant effort and resources, which limits practical use. However, our analysis of long-context vulnerabilities and the factors affecting model susceptibility suggests possibilities for more efficient attack methods. Based on the findings from Section 4, we explore simpler alternative approaches. We specifically focus on Llama models, which showed lower ASR in previous experiments, to demonstrate comparable effectiveness with reduced implementation complexity. The detailed results for the Qwen models are in Appendix D.

5.1 Fake Data Attack

One of the simplest ways to fill context is using fake data, including meaningless word combinations or synthetic text that requires minimal effort to generate. Experiments demonstrate that shot characteristics such as harmfulness and topics have minimal impact on vulnerability patterns, suggesting that the success of these attacks might be completely independent of the semantic properties of the examples.

Following these findings, we hypothesize that fake data could serve as an efficient alternative to carefully crafted examples. To test this hypothesis, we conduct experiments using *Fake-512* and *Fake-Text* that require minimal effort to generate yet may still exploit these inherent model weaknesses.